

Boss Level

Homework: The Four Moves

Name:

Date:

No computer needed! Show what you can do with an array and a loop. Finish by designing a program of your own.

Part 1 — Match the Move

Write the letter of what each move does in the box next to its name.

- A. Adds every item together into one number.
- B. Finds the largest item in the list.
- C. Counts how many items pass a test.
- D. Checks whether a value is somewhere in the list.
- E. A variable made before the loop that remembers a running answer.
- F. A boolean that starts false and flips to true when you find something.

Total

Biggest

Count

Search

Accumulator

Flag

Part 2 — True or False

Write T (true) or F (false) in each box.

An array with 6 items has a last index of 6.

The loop condition should be `i < list.length`, not `i <= list.length`

Your total variable should be created before the loop starts.

Count and Total use the same loop shape — only the line inside changes.

When searching, the flag should start as true.

Part 3 — Predict the Output

Use this array for all of Part 3:

The array:

```
const ages: number[] = [12, 7, 15, 9];
```

A:

```
console.log(ages.length);  
console.log(ages[0]);  
console.log(ages[3]);
```

Prints (three lines):

B:

```
let total: number = 0;  
for (let i = 0; i < ages.length; i++) {  
  total = total + ages[i];  
}  
console.log(total);
```

Prints:

C:

```
let count: number = 0;  
for (let i = 0; i < ages.length; i++) {  
  if (ages[i] >= 12) {  
    count = count + 1;  
  }  
}  
console.log(count);
```

Prints:

Part 4 — Finish the Code

Fill in the missing pieces. Use the ages array from Part 3.

Find the *biggest age*:

```
let highest: number = ages[____];  
  
for (let i = 0; i < ages.____; i++) {  
  if (ages[i] ____ highest) {  
    highest = ____;  
  }  
}
```

Write the four missing pieces in order:

Part 5 — Spot the Error

This program should count how many ages are under 10, but it always prints 0. Why?

The broken code:

```
for (let i = 0; i < ages.length; i++) {  
  let count: number = 0;  
  if (ages[i] < 10) {  
    count = count + 1;  
  }  
}  
console.log(count);
```

What is wrong, and how would you fix it?

Part 6 — Design Your Own

Invent a small program that uses an array and ONE of the four moves.

What is your array about, and which move will you use?

Write your program here (it's okay if it isn't perfect!):